

Atharva Waingankar

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PROFESSIONAL SUMMARY

Machine Learning Engineer with hands-on experience building end-to-end ML pipelines, deep learning models, and production inference systems. Graduate student at NC State (4.0 GPA) with published research in deep learning and applied work spanning NLP, time-series forecasting, and anomaly detection.

EDUCATION

North Carolina State University

Master of Computer Science

• **GPA:** 4.0/4.0

• **Coursework:** Design and Analysis of Algorithms, Software Engineering, Automated Learning and Data Analysis, Object Oriented Design and Development, Internet Protocols, Neural Networks

May 2027

Raleigh, NC

Mukesh Patel School of Technology Management & Engineering

Bachelor of Technology, Information Technology

• **GPA:** 3.45/4.0

May 2025

Mumbai, India

EXPERIENCE

Ultimate Scaler Pvt. Ltd.

Product Development Intern

Dec 2024 - May 2025

- Built Python-based data pipelines to ingest and process real-time financial market data through KiteConnect APIs.
- Engineered features using RSI, EMA, ATR, Bollinger Bands, and Supertrend indicators to support quantitative trading strategies.
- Assessed automated trading strategies through historical backtesting to inform algorithmic execution decisions.

PROJECTS

Solar Energy Generation Prediction using Deep Learning

- Developed and compared LSTM, GRU, and MLP models in TensorFlow to forecast photovoltaic energy generation using approximately 13,000 historical observations.
- Built an end-to-end preprocessing pipeline covering cleaning, normalization, feature selection, and hyperparameter optimization.
- Reduced feature space from 28 to 15 features using Sequential Forward Selection, maintaining predictive performance.
- Achieved best performance with an MLP model (MAE: 0.0244) and published findings at SETCOM 2025.

LeakLens: Real-Time Secret Leak Detection Tool

- Built a Go-based security platform using rule-based detection and BERT inference to identify API keys, JWTs, and SSH keys.
- Designed an AI inference pipeline to classify sensitive data from videos and images using OCR and transformer models.
- Developed secure REST APIs and unit tests to support reliable deployment and scalable inference.
- Implemented confidence scoring and automatic redaction to reduce false positives and improve detection accuracy.

Kairos

- Designed an event-driven trading simulator for market data ingestion, portfolio simulation, and automated order execution.
- Built an anomaly detection pipeline to identify abnormal market behavior using statistical and machine learning techniques.
- Integrated REST APIs and WebSocket communication to support real-time inference and live trading dashboards.

Multilingual Sentiment Analysis

- Developed a Gujarati sentiment classification pipeline using TF-IDF vectorization and supervised machine learning.
- Applied SMOTE oversampling to address class imbalance and improve model robustness across minority classes.
- Evaluated multiple classifiers using precision, recall, F1-score, and confusion matrices.

Expertiza: Open Source Contribution

- Contributed production-quality Ruby on Rails code to the Expertiza open-source software engineering platform.
- Developed 200+ automated RSpec tests validating authentication, authorization, and controller behavior.

SKILLS

- **Machine Learning:** TensorFlow, Scikit-learn, LSTM, GRU, BERT, Feature Engineering, Hyperparameter Optimization, Model Evaluation
- **Data Science:** Pandas, NumPy, Matplotlib, Seaborn
- **Languages:** Python, SQL, Go, Java, JavaScript
- **Frameworks:** Node.js, Express.js, React.js
- **DevOps & Tools:** Git, GitHub Actions, CI/CD Pipeline
- **Databases:** MongoDB, SQLite